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Investigating the use of design thinking in identifying wicked problems of startups

Parisa Jourabchi Amirkhizi[1], Farshad Zeyghami[2] and Morteza Pourmohamadi[3*]

*Correspondence: Morteza.Pourmohamadi@sydney.edu.au

[1] Tabriz Islamic Art University, Tabriz, Iran
[2] Designer at Dynamic Studio, Tehran, Iran
[3] Lecturer in Design at The University of Sydney, Sydney, Australia

Abstract

Design thinking is defined as using strategies from the field of design to solve problems in the disciplines that are traditionally not considered design. Design is a way of dealing with issues that do not have a unique and decisive solution as scientific problems do and that makes it a good approach for solving wicked problems. This research explores the utilization of design thinking in identifying and refining the wicked problems that often precipitate failure. Over 18 months, we examined five startups applying the Research through Design (RtD) method and Double Diamond framework (DDF) in 4 discovery, define, develop, and deliver stages, to reveal the underlying issues obscured by surface-level problem identifications. Our findings demonstrate a notable misalignment between startup offerings and market requirements, predominantly due to the misdefinition of problems applying design thinking. Startups tend to simplify complex, wicked problems and short-circuit the essential problem of space exploration. Results indicate that one of the most critical factors of insufficient attention to the problem is the superficial analysis of three factors of business development (i.e., human desirability, business viability, and technology feasibility) and also lack of recognition of wicked problems. We conclude that training startup businesses with the vital skills mentioned can reduce their failure rate.

Keywords: Design thinking, Business startup, Wicked problem

Introduction

Startups, from their inception, grapple with a myriad of challenges, ranging from fierce competition and rapid technological changes to the constant need for innovation (Crnogaj & Rus, 2023). A significant percentage of startups face hurdles early on, leading to a high rate of failure within the initial stages of their journey (Chincholkar, 2021; Griva et al., 2023). In the face of challenges, a recurring cause of entrepreneurial failure is the mismatch between startups' offerings and market needs (Eisenmann, 2021; Sanasi et al., 2023). As Forbes echoes, this misalignment remains a prevalent issue, contributing to the early setbacks of many ventures (Fig. 1) (CB Insights, 2024).

In response to these complex, multifaceted challenges, often described as 'wicked problems' due to their persistent resistance to resolution, design thinking has been adopted by startups as a beacon of hope (Cankurtaran & Beverland, 2020; Micheli et al., 2019; Ney et al., 2019). This methodology is characterized by its iterative nature and

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Fig. 1 Top reasons startups fail (CB Insights, 2024)

empathic approach, celebrated for spearheading innovation through continuous feedback (Razzouk & Shute, 2012; You, 2022). Evident in the lauding by esteemed journals and economic publications, the imprints of design thinking's importance are clear, its impact growing ever more prevalent in the world of entrepreneurship (Cousins, 2018; Dunne, 2018; Sarooghi et al., 2019).

Open Innovation 2.0 complements design thinking principles by promoting ecosystem-based innovation. This theory emphasizes open collaboration among businesses, governments, and academia. By tapping into the collective intelligence and resources of diverse stakeholders, startups can improve problem-solving and drive sustainable innovation (Curley & Salmelin, 2017).

The Complex Systems and Innovation Theory by Uhl-Bien and Arena (2017) offers a framework for understanding dynamic interactions in complex systems. It highlights the role of adaptive leadership and collaborative networks in promoting innovation. The theory suggests startups must engage with their environments and continually adapt their solutions to address complexities effectively.

Change Management in Innovative Organizations, as discussed by Anderson and Anderson (2010), focuses on the strategies and processes required to manage change within dynamic and innovative environments. This theory highlights the importance of creating a shared vision, fostering a culture of innovation, and empowering employees to take initiative. Design thinking facilitates these processes by promoting a user-centric approach and iterative development, enabling startups to navigate through change more effectively.

The efficacy of design thinking spans across industries, including technology, healthcare, and education, showcasing its adaptability and impact in a wide range of startup landscapes (Lewrick, 2022; Lynch et al., 2021; Oliveira et al., 2024). A meta-analysis by Bhandari (2022) reinforces this notion, establishing a linkage between design thinking and a 60% improvement in critical development processes, such as rapid prototyping and iterative testing. These insights are further substantiated by research from Carlgren et al., 2016a, 2016b, 2016c), which, with its large data set, endorses the findings of prior longitudinal studies and underscores the need for industry-wide analyses.

Design thinking has emerged as a valuable approach for identifying and addressing complex problems faced by startups. These problems, known as wicked problems, are characterized by their complexity and uncertainty, requiring innovative solutions that go beyond traditional problem-solving methods (Bartoloni et al., 2022; Head, 2022). Design thinking offers a multidisciplinary approach that focuses on desirability, viability, and feasibility, aligning human needs, business needs, and technical needs (Bertoni & Askling, 2024). By integrating design thinking into the enterprise, organizations can effectively tackle complex problems related to strategy and innovation (Cai et al., 2023; Campbell & Carayannis, 2016).

Startups, operating in environments of extreme uncertainty, can benefit from the application of design thinking to navigate through wicked problems (Thoring & Mueller, 2023). Design thinking fosters innovation by generating concepts for new products and services while developing solutions for complex challenges (Rösch et al., 2023). By adopting design thinking, social enterprises can effectively address wicked problems, showcasing their commitment to innovation and problem-solving (Pira et al., 2024).

The application of design thinking in startup business planning involves stages, such as generalizing startup and design thinking concepts, identifying startup life cycle stages, and analyzing design thinking capabilities within startups (Heryani et al., 2024). Design thinking, with its emphasis on user interaction and value creation, aligns well with the lean startup philosophy, making it a suitable approach for entrepreneurial teams

addressing complex problems (Bender-Salazar, 2023). Furthermore, design thinking's user-centered innovation approach, revolving around inspiration, ideation, and implementation, provides a structured process for addressing wicked problems (Tsai et al., 2023). Therefore, design thinking serves as a powerful tool for startups to identify and tackle complex problems by fostering innovation, engaging users, and creating value in a dynamic and uncertain business landscape. By integrating design thinking into their strategic decision-making processes, startups can navigate through challenges and drive sustainable growth.

As startups march into the digital era, they encounter new terrains reshaped by AI and analytical tools, with design thinking evolving to meet these advancements head-on (Baradaran, 2019; Dell'Era et al., 2020). Scholars such as Pham et al. (2022) suggest that this new era demands an overhaul of design thinking principles to emphasize data-centric strategies. This perspective coincides with the call for creating adaptable, resilient startup cultures, another aspect, where design thinking plays a crucial role, yet is often underestimated (Govindan et al., 2020; Halpern & Walther, 2022; Linton & Klinton, 2019).

Originating from creative disciplines, design thinking is a potent framework for addressing complex problems (Dorst, 2011). Guided by principles such as empathy and iterative learning, it promotes collaboration across disciplines (Razzouk & Shute, 2012). Herbert Simon's "The Sciences of the Artificial" laid the groundwork, advocating for problem-solving akin to design processes (Zhou, 2021). This approach facilitates deep engagement with complex issues, crucial for tackling wicked problems. By aligning research with design thinking, we highlight its relevance in addressing misalignments between startup offerings and market needs in entrepreneurship.

Despite its well-documented benefits, a paradox exists, while design thinking is acknowledged as a tool for dissecting and diagnosing problems within startups, its role in identifying and confronting these wicked problems before they escalate has not been sufficiently explored (Huang & Hands, 2022; Moore, 2020). Therein lies our research gap: to pivot from using RtD as a mere solution-provider to a diagnostic apparatus, and to understand what it reveals about the complexities faced by startups (Kimbell, 2011; Stappers & Giaccardi, 2017). This study thus raises pivotal questions:

> RQ1: Why do startups present products or services that do not match the market needs?
> RQ2: How can RtD, within the realm of design thinking, aid in identifying and articulating these misalignments?
> RQ3: What does design thinking reveal about the identification and articulation of wicked problems faced by startups?

By addressing these questions, we confront the issue of principle–practice disconnection, which may hinder the full realization of factors of business development: desirability, viability, and technological feasibility (Nesner et al., 2020; Staniec & Pilawa, 2020). Aiming to close these gaps, this paper not only reiterates the value of design thinking in navigating the uncertainties that startups face but also provides empirical data supporting its effectiveness. By showcasing 18 months of application in five startups, we: (1)

solidify the study's critical nature and current relevance by highlighting the disconnects between design thinking ideals and entrepreneurial practices; (2) emphasize our unique research niche challenging the notion that design thinking is not empirically grounded; and (3) endeavor to decode the strategic use of design thinking as more than just a creative tool but as a strategic navigator through entrepreneurial complexity.

In the following, a concise yet comprehensive review of design thinking in the business world, and explore the intricacies of wicked problems. In the methodology section, an overview of RtD and the DDF is provided, which is followed by a summary of the process and steps employed throughout the study. This section is also complemented by a graphical abstract that encapsulates the entire research plan. Finally, the results, discussion, and conclusion segments elucidate the principal findings and outcomes of our research endeavors. To measure the impact of design thinking on business performance, we incorporated the key performance indicator of revenue Growth (RG). This metric allows us to quantitatively assess the financial growth of startups before and after the application of design thinking methodologies.

Design thinking and wicked problems

Design thinking's evolution from a method used in design and architecture to a broader problem-solving approach underscores its significance in contemporary innovation (Pressman, 2018). Central to this approach is the iterative cycle of empathy, ideation, prototyping, and testing, a methodology that embraces complexity and champions human-centric solutions (Razzouk & Shute, 2012). This framework's adaptability to wicked problems, issues characterized by their nonlinear, unpredictable nature, highlights its potential to transform entrepreneurial strategies.

Design thinking has emerged as a transformative paradigm for addressing complex and ambiguous challenges across various domains. Its evolution from early twentieth-century design principles to a comprehensive problem-solving methodology underscores its significance in contemporary innovation and educational landscapes. Scholars such as Horst Rittel have been instrumental in developing the frameworks that form the basis of design thinking, especially in tackling 'wicked' problems, complex issues that defy straightforward solutions (Cohen & Prosina, 2020; Cross, 2006; Rittel & Webber, 1973).

Design thinking, a methodology grounded in the core principles of empathy, ideation, and experimentation, has emerged as a transformative approach to addressing complex and ambiguous challenges across diverse domains (Jacobs, 2018). Originating from the fields of industrial design and architecture, design thinking has evolved to embrace a problem-solving ethos that values human-centered design, iterative learning, and interdisciplinary collaboration (Pimental, 2023). The iterative nature of design thinking, characterized by a continuous cycle of understanding, ideation, and solution testing, marks a departure from linear problem-solving methods (Elsbach & Stigliani, 2018; Micheli et al., 2019). This approach is vital in addressing the uncertainties and complexities inherent in wicked problems, with scholars such as Pasisi (2014), Plattner (2015), and Proczek and Garbarczyk (2023) highlighting its efficacy through a human-centered lens.

Recent literature has focused on the integration of design thinking with emerging technologies and its application beyond traditional domains. Earle and Leyva-de la

Hiz (2020) propose a conceptual model that merges augmented and virtual reality with design thinking to enhance sustainability-focused education. Similarly, Cooke et al. (2020) advocate for incorporating design thinking processes into game design frameworks to bolster problem-solving skills among students, particularly in navigating wicked problems. Matthews et al. (2022) extend the discussion to co-design methodologies, emphasizing their potential in education and social justice arenas.

The competitive dynamics of innovation ecosystems, as explored by Brusoni and Prencipe (2013), underscore the relevance of design thinking in coordinating knowledge requirements effectively amidst uncertainty and complexity. This highlights the approach's adaptability and its critical role in shaping the capabilities of focal firms within these ecosystems.

Furthermore, the application of design thinking in future forecasting and addressing societal challenges, such as migration and refugee issues, illustrates its breadth (Proczek & Garbarczyk, 2023; Sanzeni et al., 2019). The emphasis on empathy, collaboration, and experimentation within design thinking offers a framework for developing holistic and adaptive solutions to wicked problems (Chan & Xiang, 2022; Shen, 2023).

Acklin and Wanner (2017) point out the need for enhanced collaboration between technical and design professionals, especially in high-tech startups, suggesting that the operationalization of design thinking in modern contexts may require further exploration and adaptation to contemporary challenges.

Wicked problems of businesses

Wicked problems, initially coined by Rittel and Webber, encapsulate the entangled, continually evolving issues that defy linear, analytical approaches traditionally employed for 'tame' problems. These problems are embedded in societal complexities, lacking clarity, definitive solutions, and often exacerbating as attempts to solve them lead to contention among stakeholders with divergent viewpoints (Rittel & Webber, 1973).

Urban planners first identified wicked problems in the context of social policy and planning (Head & Alford, 2015; Hull et al., 2020). Wicked problems remain dynamic, compounded by moral and political dimensions, demanding a methodological shift in complex societal and organizational problem-solving. Amidst the quandaries of the digital era and global interconnectedness, wicked problems in business are becoming more prevalent, adding layers to the complexity that organizations must navigate (Camillus, 2008; Pierre, 2017).

As Rittel and Weber suggested, planning and design, guided by innovation in response to complex client briefs, cannot adopt the problem-solving strategies of scientists or engineers (Rittel & Webber, 1973). The wickedness of these problems, with their inherent indeterminate nature and competing values, impacts both their understanding and potential solutions. Wicked problems command a problem space exploration, requiring approaches that dissect and redefine the issues at hand, grounded in human-centered design; namely, human needs, technological possibilities, and business acumen (Kolko, 2011).

In the context of startups, misidentifying the problem, either through focusing on solution spaces, overlooking critical factors of business development, or failing to differentiate between concern and the actual problem, can bear dire consequences. It is

incumbent on these businesses to rigorously define problems by genuinely considering the entire spectrum of factors: human desires, technological feasibilities, and business viability, moving beyond isolated components to a holistic view of the intricate web of wicked problems.

The wicked problems faced by businesses in the twenty-first century are complex and challenging, requiring a different approach to traditional strategic planning processes. The research conducted by Latesteijn and Rabbinge (2012) highlights the challenges faced by the Dutch innovation program TransForum in developing sustainable modes of agricultural production. The study emphasizes the need for practical wisdom in decision-making to address wicked problems effectively. Intezari and Pauleen (2018) argue that conventional reductive approaches are limited in solving complex problems and suggest incorporating wisdom into management decision-making. The authors propose a model based on empirical findings to identify qualities and processes involved in making wise decisions, especially in management. Devins and Jones (2016) discuss the challenges faced by family businesses, particularly in succession planning, which require a different approach due to the complexity of wicked problems. The authors present a framework to engage multiple stakeholders and reconcile family/business interests for sustainable family business futures. Moses and Vest (2010) examine the actions of Coca-Cola and PepsiCo in South Africa, highlighting the challenges of social and ethical responsibilities in business decision-making. Mouritz et al. (2022) provide insights from running a Master's unit on Leadership in Sustainability, emphasizing the importance of collective wisdom, conversations, creativity, contemplation, and courage in addressing wicked problems in small business enterprises. Minatogawa et al. (2022) identify critical gaps in the literature. The study emphasizes the need to understand SBMI as a wicked problem and explores the role of dynamic capabilities, open innovation, and tools supporting SBMI[1] implementation for sustainable development. Overall, the literature highlights the importance of practical wisdom, stakeholder engagement, and innovative approaches in addressing the wicked problems faced by businesses in the modern era. Researchers can benefit from these insights to develop strategies for managing complex challenges in business environments.

Design thinking and business

Certainly, the impact of technology, including in the realms of entrepreneurship and business, has been extensively scrutinized (Carayannis et al., 2015). John E. Arnold's foresight in combining human values with technical and business innovation (Fig. 2) is as pertinent now as it was at the inception, guiding the development of holistic solutions well-suited for our fast-evolving digital and business landscapes (Arnold, 1959). In the contemporary analysis of business practices, especially in relation to the burgeoning startup culture, the application of Arnold's principles in design thinking is evident. The integration of design thinking into business, particularly within startups, echoes Arnold's early concepts, marking a key point of convergence among various studies. Açar and Rother (2011) explore the practical realization of design thinking within engineering

[1] Sustainable Business Model Innovation.

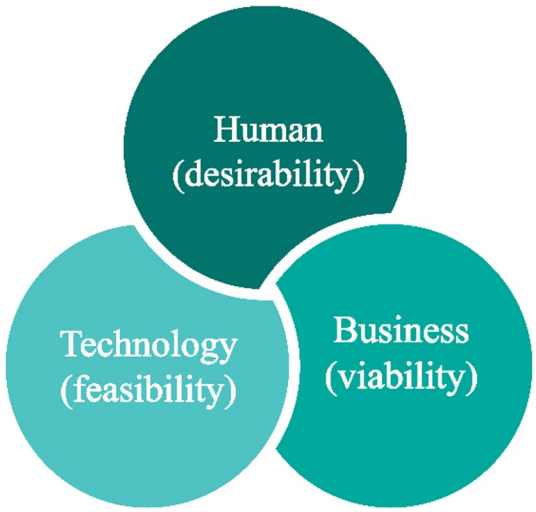

Fig. 2 Three factors of business development

education and consequentially, technology startups, acknowledging this methodology as a catalyst for competitive advantage, a sentiment in line with Arnold's interdisciplinary approach. Similarly, Mansoori and Lackeus (2020) highlight how design thinking can significantly impact a startup's market penetration, risk management, and success predictability in lean startups.

Diving deeper, design thinking acts as a bridge between innovative ideation and practical business results (Nakata & Hwang, 2020; Tschimmel, 2012), resonating with Arnold's focus on human-centered desirability and the integration of technology in design. Beyond individual startups, the systematic integration of design thinking into entrepreneurial support systems such as incubators and accelerators plays a crucial role in fostering sustainable business success. As recent literature suggests, this structured approach enhances problem-solving capabilities and market alignment. The integration of design thinking into educational programs, incubators, and accelerators is essential for fostering successful startups by emphasizing empathy, ideation, prototyping, and testing (Sadiku et al., 2019). Despite its significance, the concept remains ambiguously defined, necessitating clearer frameworks in entrepreneurial education. Kwon et al. (2024) highlight its role in social enterprise growth, demonstrating its impact across diverse cultural contexts. Vnukova et al. (2022) further argue that design thinking enhances startup success by prioritizing human-centered solutions. Incubators and accelerators play a crucial role in supporting startups, as participation in these programs correlates with improved revenue, job creation, and funding (Ayatse et al., 2017; Lange & Johnston, 2020). Dell'Era et al. (2020) emphasize that design is not merely aesthetic but a strategic innovation tool, while Dobrigkeit et al. (2020) advocate combining design thinking with lean startup methodologies for improved product development. The educational implications are significant, as Tickoo and Grammer (2022) highlight the gap between academic training and workforce needs, which can be bridged by incorporating design thinking into curricula. Carlgren et al., 2016a, 2016b, 2016c) reinforce its role in fostering innovation across sectors, including education. Ultimately, integrating design thinking into entrepreneurial

frameworks enhances problem-solving, promotes innovation, and leads to more user-centered and sustainable business solutions. These findings underscore the necessity of embedding design thinking principles not only within individual startups but also at the ecosystem level, influencing the way entrepreneurial environments are structured and developed. Currently, there is a focus on the design, establishment, organization, and evolution of entrepreneurial ecosystems (Alvarez et al., 2018). These studies also echo the need for further research in the field to substantiate the scalability and effectiveness of design thinking across varied business sectors (Kerguenne et al., 2023). Design thinking is even applied to address how small- and medium-sized manufacturing enterprises can effectively implement Artificial Intelligence, exploring its potential to create innovative (service) business models in the face of global competition (Garrel & Jahn, 2022).

Building on the scholarly work, the potential of design thinking in larger organizational structures is contemplated (Ney et al., 2019; Vinnakota & Narayana, 2014) even within the domain of organizational employee training (Woraphiphat & Roopsuwankun, 2023). The propositions put forth by these researches not only advocate for the method's applicability in complex corporate settings, as originally proposed by Arnold but also encourage more interdisciplinary research that is aimed at studying the actual applications and real-world implications of design thinking. Though initiated by Arnold, it is through ongoing implementation and education by institutions such as Stanford and corporations like IDEO, that his vision for merging human values with technology and business innovation has been fully realized, forming the bedrock of current entrepreneurial and product development practices (Gonen, 2020; You, 2022).

Integrating design thinking into startups

In the fast-paced startup world, integrating design thinking into strategic and operational processes is vital for innovation and sustainability (Olorunfemi, 2024). This user-centric approach helps navigate uncertainties, enhancing decision-making, product development, and team collaboration (Mahmoud-Jouini et al., 2019; Naglinska, 2021). This section explores how systematically incorporating design thinking at strategic and operational levels can improve decision-making, product development, and team collaboration. Integrating design thinking begins with its alignment to the startup's core vision and mission (Sarooghi et al., 2019). This alignment ensures that every strategic initiative and operational activity undertaken is deeply rooted in understanding user needs and delivering innovative solutions. Startups, therefore, should embed design thinking principles, empathy, ideation, prototyping, and testing, into their vision and mission statements (Bodini, 2020). This incorporation not only signals a commitment to a user-centered approach but also sets a foundational ethos for the organizational culture and strategic direction (Prus, 2018).

At the strategic level, design thinking offers a robust framework for decision-making by encouraging a bias towards action, a culture of experimentation, and an openness to learning and iterating (Carlgren et al., 2014; Zheng, 2018). Startups can employ design thinking methodologies to evaluate market opportunities, develop business models (Müller & Thoring, 2012), and strategize entry points into competitive landscapes (Sarooghi et al., 2019). Through such frameworks, startups can enhance their agility,

making informed pivots and adaptations based on continuous feedback from real-world interactions with their target market.

Operationally, the application of design thinking transforms the product development cycle (Carlgren et al., 2016a, 2016b, 2016c; Carlgren & BenMahmoud-Jouini, 2022) . By adopting an iterative process that champions rapid prototyping and user feedback, startups can significantly reduce the time and resources typically required to bring a new product to market (Edison et al., 2018). This process begins with empathizing with the user to deeply understand their needs, followed by ideating solutions that meet these needs creatively (Fuchs et al., 2019). Prototyping and testing these solutions with users then informs refinements, ensuring that the final product is both innovative and closely aligned with user expectations.

Design thinking fosters a culture of collaboration and continuous learning within startup teams (Elsbach & Stigliani, 2018; Snyder et al., 2018). By breaking down silos and encouraging cross-functional teams to work together in solving problems, design thinking facilitates a blend of diverse perspectives and expertise (Kaihua, 2019). This collaborative environment supports a culture, where failure is seen as an opportunity for learning and growth, essential for the fast-paced and often unpredictable startup ecosystem.

For startups, where resources are invariably limited, design thinking provides a framework for more effective allocation and prioritization (Leifer & Meinel, 2019; Ney & Meinel, 2019). Through its iterative cycles, startups can quickly identify which features or aspects of their product resonate most with users, allowing them to allocate resources more efficiently towards developing these features. This prioritization ensures that startups are not only managing their resources wisely but are also continually enhancing user satisfaction and loyalty. As part of our structured approach to exploring the influence of design thinking, we have formulated the following hypotheses. These are derived from our initial research questions and are intended to provide a focused framework for our investigation:

H1 Design thinking significantly improves the ability of startups to accurately identify wicked problems.

H2 Startups that employ design thinking have higher success rates in managing wicked problems compared to those that do not.

H3 Design thinking enhances the alignment between startup offerings and market needs.

Materials and methods

The philosophical approach of this study is anchored in emphasizing practical problem-solving and user experience, which aligns perfectly with both the RtD method and DDF. The RtD method enables in-depth exploration of the design process for design challenges while acknowledging the need for clear quality indicators (Prochner & Godin, 2022). RtD is an approach to conducting scientific research that uses

the methods, techniques, and processes of design practice to generate new knowledge and advocates for a practical and action-oriented approach to research problems. (Zimmerman & Forlizzi, 2014). The RtD method was elected specifically for its strength in dealing with ill-structured, 'wicked problems' (Ylirisku et al., 2016), a common characteristic of challenges faced by startups (Bender-Salazar, 2023). RtD approach differs significantly from traditional scientific research methods in a way that it uses design power as a reflective action to continually reinterpret and reformat a problematic situation through the process of building and reviewing products or services that act as proposed solutions (Philippi, 2018). The method promotes the creation of solutions through direct engagement with real-world challenges and their associated constraints (Maher et al., 2018). RtD is a systematic method in the process of interpreting and reinterpreting the conventional understanding of the world and generally requires accurate and logical documentation of the actions taken during the design process. The most important distinction between RtD and design practice is the intent that design researchers apply to a problematic situation.

At RtD, design researchers focus on how new and valuable knowledge is generated by design practices. Therefore, this method will be the best method to use in the present study. RtD can be a good way to talk about things we need to know more about or gain a deeper understanding of. In short, RtD helps to better understanding of the problem, developing research tools, and discovering the deeper and more realistic needs that are required in this research. On the other hand, it should be noted that business issues with a human audience are generally considered wicked problems (Ghauri et al., 2020). One way to solve ill-structured problems is trial and error. Research through design is the closest and most appropriate method that by testing solutions from the beginning, helps to use the right research tools, to find the right problem and ultimately the right solution to these problems faster and more effectively (Gaver, 2012). The DDF, on the other hand, provides a structured process model that aligns with the RtD approach (Svensson, 2023).

In the following, the DDF design framework was used as a design framework. The illustration in Fig. 3 outlines the overall progression of the present study. The reason for choosing this framework is the strong theoretical basis of it, which provides a suitable platform for studying business problems. In addition, the separation of problem and solution space in DDF is consistent with the focus of this project, which is the two main concepts of "problem" and "solution" (Gustafsson, 2019). This framework (Fig. 4) is one of the famous design frameworks, enables us to begin with a broad exploration of the problem space through divergent thinking (Tschimmel, 2012) which has four main stages:

- Discover: Discovering the user deep requirements through divergent thinking
- Define: Defining the problem through convergent thinking
- Develop: Development of possible solutions through divergent thinking
- Deliver: Testing solutions, implementation, and feedback through convergent thinking

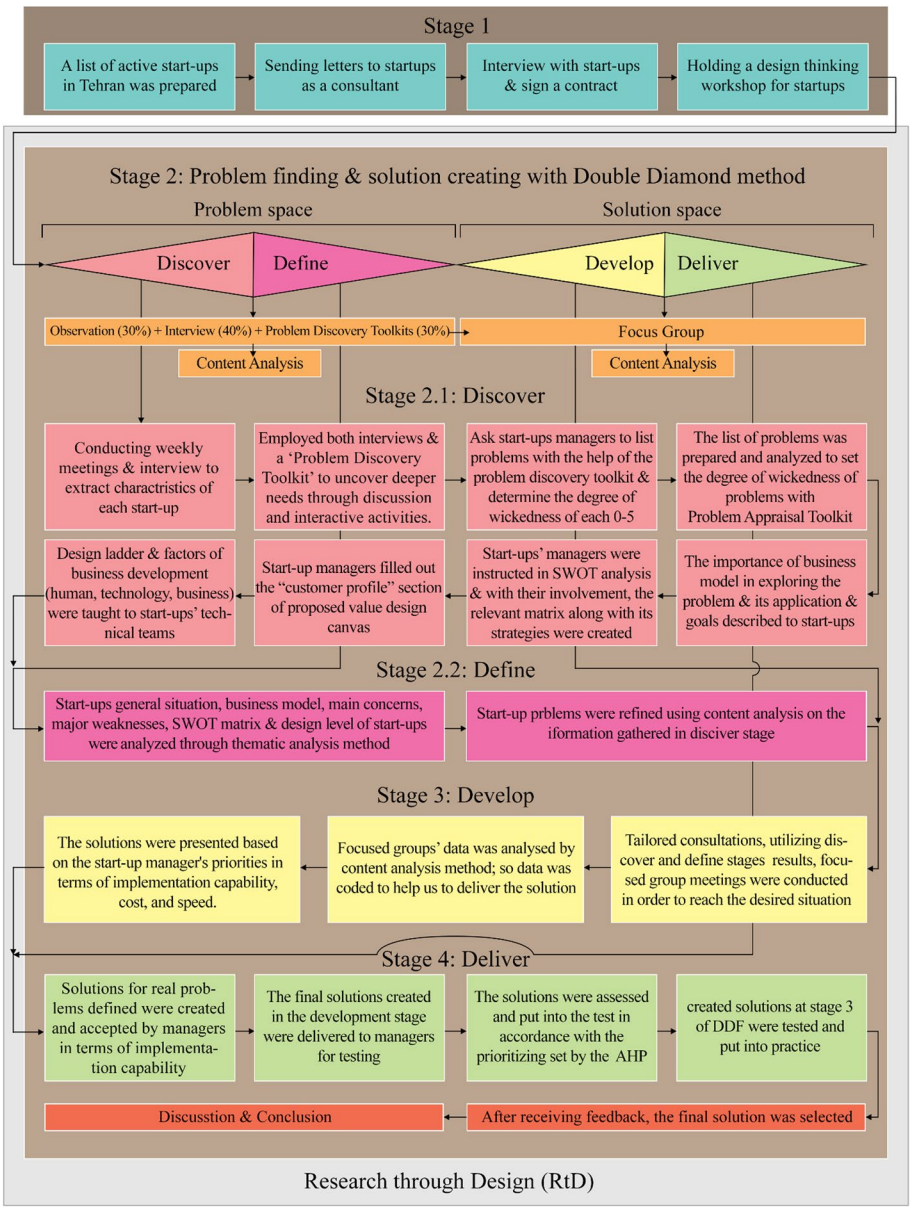

Fig. 3 Research framework

There are three main components in DDF. 1. Concern: general challenge or designer's need to create a favourable situation. 2. Problem: the exact definition of the problem. 3. Solution: the exact definition of the solution. There are also two main spaces in this framework. 1. The problem space: finding the right problem, and 2. the solution space: finding the right solution. The purpose of the DDF is to turn the general challenge (concern) into a practical solution. In this framework, we are faced with two different types of thinking regarding problems and solutions, which tell us how to act at each stage:

- Divergent thinking: broadly thinking, keeping an open mind, considering anything.
- Convergent thinking: think hard enough, focus and identify one or two main problems and solutions (Fig. 4).

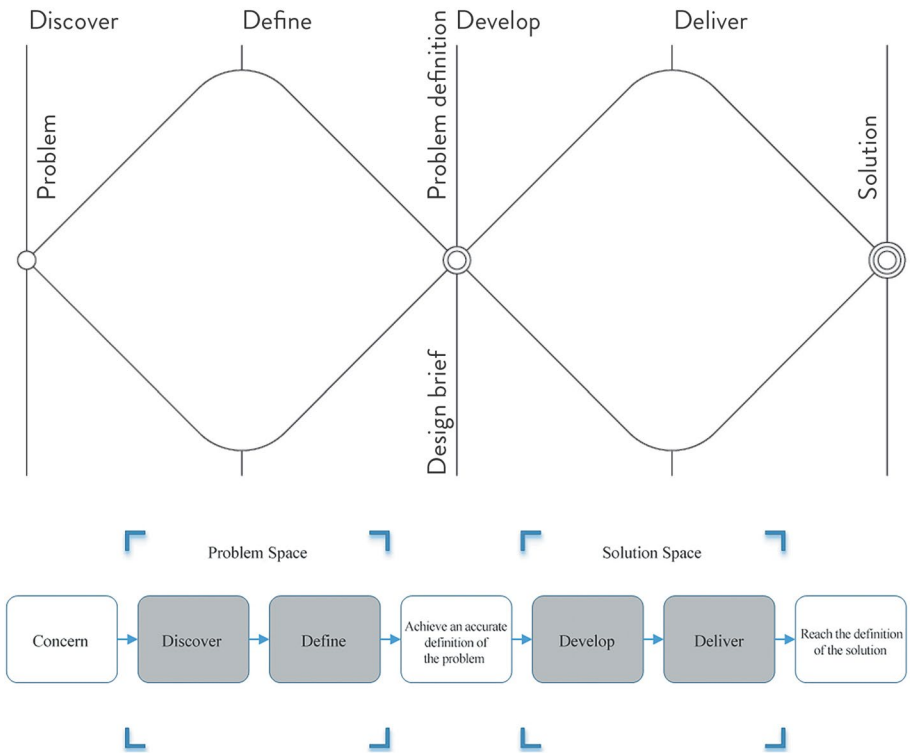

Fig. 4 Research method in the framework of the DDF

Startups increasingly employ RtD and DDF to enhance their business models and services, driving innovation. McDonald & Eisenhardt (2020) introduced the concept of parallel play, enabling entrepreneurs to accelerate learning and create value. Xie et al. (2023) applied DDF to design a mixed-reality navigation system, improving customer experiences in the tourism industry.

Moreover, startups utilize DDF alongside other methodologies, such as design thinking, lean thinking, and process management (Caulliraux, 2020). Adaptations in healthcare include an agile approach and PDSA[2] cycles (West, 2018). Design and design management professionalize startups and enhance access to network resources (Acklin, 2017). In addition, Xie et al. (2023) provided a model for accelerator managers' decision-making, combining design thinking methods with quantitative analysis for optimal startup selection.

The study was conducted in Tehran, the capital city of Iran, chosen strategically for its rich concentration of business accelerators. Being the political and economic hub, Tehran not only facilitates easier access to numerous accelerators but also presents a unique urban and economic landscape ideal for examining the influence of localized variables on these entities. Variables such as urban density, which impacts accessibility and networking opportunities; economic development levels, affecting resource availability and market size; and local governance policies, which influence operational environments

[2] Plan-Do-Study-Act.

Table 1 List on Tehran active startups

Startups	Business field
Avatech	General
Dmond	General
Tivan	General (Entrepreneurship)
Finnova	Financial technology (FinTech)
Trigup	General
Karmana	General with a focus on Internet of Things, Blockchain, FinTech
Sharif	General field, the team includes at least one student of Sharif University
Jahesh	General (Accelerator of Shahid Beheshti University)
Rooyesh	General (The Academic Center for Education,Culture and Research)
Shezan	Health, energy, water and environment
Farabi	FinTech
Karencrowd	General, virtual accelerator (funding system for startups and investment)

Table 2 Abbreviated names of startups in order of growth stage

Stage of growth	Startups
Pre-seed stage	A
Seed stage	B
Early stage	C
Mid stage	D
Growth stage	E

through business incentives and regulatory frameworks were considered in choosing Tehran. These factors are integrated into our analysis to contextualize the performance of accelerators within Tehran's broader socio-economic and policy environment, offering a nuanced understanding of the dynamics influencing the success or challenges faced by accelerators in the capital.

To start RtD and DDF, a list of active startups in Tehran was prepared (Table 1). The criterion for being active is the amount of content production on the website, Telegram channel, and LinkedIn, referring others to cyberspace (through monitoring the news of startups), and the number of public and private events. The text of the proposal to cooperate as a startup consultant in solving the problem was sent to the startups, and 5 startups announced their cooperation.

Out of 12 available startups, 5-startups in five stages of growth expressed their interest to have consultation meetings with the author and were selected as case studies of the project. These startups were named as follows (Table 2) based on the stage of growth they are in due to their privacy.

Observation and interview sessions besides problem discovery toolkits, guided us through invaluable information. This triangulation reinforces the interpretation of the data. All interviews were transcribed verbatim, and data was anonymized and tabulated for content analysis. The analysis utilized the content analysis method, which serves as a research instrument for identifying specific words, themes, or concepts within qualitative data, such as text was employed. The process was initiated by coding data into

meaningful categories, which were continuously refined. Content analysis focused on identifying recurrent patterns in the data that align with RtD iterative cycles.

The coding information of the discover stage is coded to help us to define the problem. The concern of each startup gradually became a "problem" by implementing the "discover" and "define" stages and a solution was created for these "problems" by implementing the "develop" and "deliver" stages by conducting focus group sessions to reach the desired situation. The information of develop stage is coded to help us to deliver the solution.

We selected the RG as a key performance indicator to evaluate the effect of design thinking on the startups. RG provides a clear and measurable way to assess changes in financial performance, offering a direct comparison of each startup's revenue before and after the implementation of design thinking practices.

Results

Concern

The design "concern" was identified as the entry point of Double Diamond, it became a "problem" through exploration and action in the problem space, and finally it became a practical "solution" through exploration and action in the solution space (Table 3).

Discover stage results

In the first meetings, it was necessary to collect enough information about the startups, their problems and their needs. Therefore, we tried to establish a close relationship with them to find out the problem. In this regard, weekly meetings were arranged with the managers. In the first consulting session with each startup while introducing the general research ahead to the managers and explaining the basic concepts to reach a common understanding of the project goals, during an interview (user research at the speech level), the general characteristics of each startup and their RG are extracted in Table 4.

In the second session, in addition to the interview as a user research tool at the speech level, a toolkit called the problem discovery toolkit (Fig. 5) was used as a generative tool with the aim of discovering the deeper needs of the businesses at the levels of doing and making.

In the consulting session with each startup, a general explanation was given about the 4 concepts of "current situation, desired situation, problem, and solution" and the nature of the wicked problem. Startup managers were told to list their startup's problems. The results of this session is presented in Table 5.

A list of concerns was prepared and analysed with a Problem Appraisal Toolkit we designed for DDF discovery stage by consulting with 5 expert designers, which is mentioned in Table 6.

Table 3 Concerns of startups according to the founders

Startup A	Startup B	Startup C	Startup D	Startup E
How to offer a discount card	Low number of new members	Unrecognized brand	Slow production process	Low sales

Table 4 General features of startups

Startups	A	B	C	D	E
Stage	Pre-seed	Seed stage	Early stage	Mid stage	Growth stage
Category	Service—B2C	Service—B2C	Service—B2C	Product sales—B2B	Product sales—B2C
Field of wok (from public view)	Hairdresser booking with a discount	Online Forum—Social Network	Car repair on site	Designing, consulting and production of promotional gifts	Industrial kitchen appliances store
Field of work (from a professional point of view)	Platform	Open innovation platform	Two-sided platform (repairman—owner)	Customized product design	Global sourcing and kitchen layout design services
Mission	–	Crowdsourcing services based on human interactions instead of AI	On-site emergency car repair services	Development of companies' advertising through design and production of promotional gifts	High usability, physical and cognitive ergonomics
Technical platform for service/product delivery	Website and application	Website	Application	Website and social media	Showroom
Number of team members	1	4	20	4	16
Business model	Subscription	Freemium	On-Demand	Direct selling	Direct selling
Revenue growth of last quarter	$5230	$3120	$4310	$2180	$1570

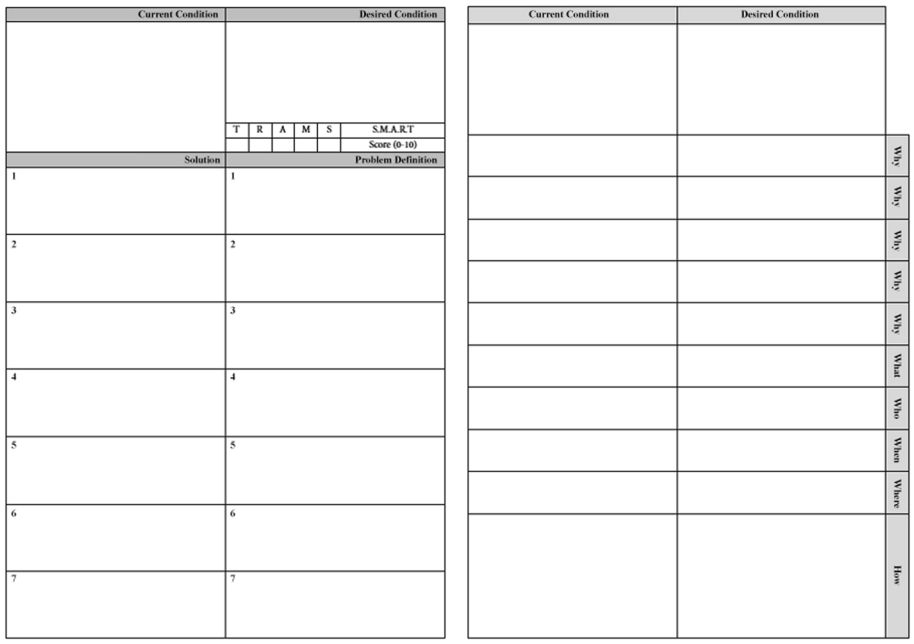

Fig. 5 Problem discovery toolkit

Table 5 Initial problem and solution from startup managers' perspective

Startups	A	B	C	D	E
Current status	A raw idea	No Interaction growth	No order growth	Slow process of production and packaging, late customer acquisition, customer loss, revenue reduction, profit reduction, loss of key personnel	Less than $5000 monthly income, debt of $40,000, delay in payment of staff salaries
Optimal status	Annual profit of $236,000	Achieve + 10/day interaction rate growth over the next 3 months	Achieve + 5/day interaction rate growth over the next 3 months	On time delivery to the customer	Debt payment, reaching a $20,000 monthly income
Problem—from the perspective of the founder	How to offer a discount card	Low new members	Unrecognized brands	Slow production process	Low sales
Solution—from the perspective of the founder	Get help with financial planning discount method	Increase capital to promote advertising	Expensive OOH advertising	Complete outsourcing of the production process	Continuous hiring of new staff for the sales department

Table 6 Problem appraisal toolkit for any startup issue

	Score (0–5)	Question (solution space)	Score (0–5)
The degree of problem clarity		The degree of solution clarity (Not necessarily the final solution)	
The problem's stability		There is a definite end point in solution	
Belonging to a similar category of problems that have the same solution		Ability to evaluate the solution correctly and wrong	
		Existence of a testable solution	
		Ability to try and error in solution	
		Existence of an alternative or suggested solution already	
Total score (the degree of problem resolution)		Total score (the degree of solution resolution)	

According to the goal of the first stage of the DDF (discovering the problem and the deep need of the user) the counselling sessions continued. In third consulting session, due to the importance of the business model and its design in exploring the problem of startups' business model's application and goals described to startups that lacked it. Then, each startup business model was extracted (Table 7).

In the fourth consulting session, the managers of the startups were instructed in SWOT analysis, and with their involvement, the relevant matrix along with its strategies were created (Table 8).

In the fifth consulting session, the startup managers filled out the "customer profile" section of the proposed value design canvas using a list of provocative questions for each section of the canvas to have the necessary information to complete the value proposition canvas in the fourth stage of design.

In the sixth session, the following topics were taught to the startups' technical teams:

Table 7 Business model of startups

startups	A	B	C	D	E
Customer section (CS)	Ladies 20–60 years	Student, specialist, business owner	Car owner	Small, medium, and large businesses	Hotel, restaurant, coffee shop, commercial complex
Value proposition (VP)	Permanent discount	Improve soft skills, reach the target community, branding and marketing	Emergency car repair on-site	Strong design based on order and business needs, reasonable price and high quality, creative design	Replacement warranty, on-site after-sales service
Customer relationship (CR)	Customer club	Support and training on website	Customer club	Website articles, high quality photos	CEO Storytelling
Channels (CH)	Application	Website, search engine, social network	Application, website, social networks	Website, search engine, social networks	Phone, sales office
Revenue (R$)	Selling discount cards	Private think tank	Percentage of sales of repair services	Product sales, consulting	Product sales, map
Key partners (KP)	Hairdressers	Student-related Businesses audiences in the digital space	Repairmen	Manufacturers of product components	Domestic equipment manufacturers
Key activities (KA)	Presentation of services to hairdressing managers	User service, user engagement	Training and justification of repairmen, maintenance of the platform	Content production for website, marketing	Marketing, sales
Key Resources (KR)	Application, the communication network of the founder of the startup	Website, active user community, team experience	Application, platform developers	Website, team experience	Managing Director, Designer, Equipment Warehouse
Cost (C$)	Marketing and sales	Workspace, server, team, marketing	Personnel rights, platform maintenance, training	Location, human capital wages, marketing, current location costs	Location costs, staff salaries, running costs, purchase of goods

Table 8 SWOT matrix results

startups	A	B	C	D	E
S	High motivation of the founder	Robust technical team	A strong investor	Creatively design and high-quality products	The unique personality of the CEO in customer attraction
W	1. Lack of knowledge, experience, team 2. Inability to build a team	Limited knowledge and skills in design within the team	Limited design knowledge and skills within the team, resulting in a weak app	Limited management knowledge & skill (human, financial, business)	1. No management knowledge in the organization 2. high debt
O	Strong manager network	1. Strong access to the academic community 2. driving organizations towards open innovation	The absence of a serious competitor	No competitors at the B2B scale, presence of loyal customers	Increase in the exchange rate and the impracticality of importing foreign goods
T	No market demand for the beauty salon discount card	1. No demand for an open innovation platform 2. high competition	Customer dissatisfaction with repair services	1. Inexpensive Chinese products 2. increase of small competitors	Growth in the number of professional competitors
SO	Using the expertise of seasoned mentors	Enhancing website features and creating rich content	Content Marketing	Content Marketing	Domestic production of equipment
ST	Content Marketing	Generating advertising content	Specialized technical training and professional behavior education for repair technicians	Creative product design and diversification in the product portfolio	CEO transitioning to Sales Manager, rebranding
WO	Consulting with Network Individuals	Employing design as a strategic approach	Hiring professional designers for UI/UX of the app	Consultation in the CRM (Customer Relationship Management)	Hiring a professional financial manager
WT	1. Collaboration with talented individuals 2. Entering a business accelerator	Elevating the level of design knowledge and skills within the team for the development of UI/UX	Establishing a User Experience (UX) Design Unit within the organization	Employment, training, or consultation in the field of strategic market management	Hiring a CEO with expertise in market strategy

- The design ladder and the need for startups to move from low levels of design to design as a strategy and use it for factors of business development.
- Three factors of business development (human, technology, business) and the importance of starting design from the human need and desire towards the other two factors

Current status of startups in terms of the level of design in startups and the amount of attention they pay to business development factors was extracted through a focus group meeting with the startup's technical team (Table 9).

All the data and information obtained from the discover stage from all 5 startups, through observation and interview sessions besides problem discovery toolkits, were examined using the content analysis method. The coding information of discover stage coded to help us to define the problem (Table 10).

Define stage results

In the discover stage, the general situation of startups, business model, main concerns, major weaknesses, SWOT matrix and design level of each startup were determined. Design thinking is based on creative solutions and a multi-faceted approach to problems, so the emphasis on finding the right problem is more than the emphasis on finding the right solution, since the choice of problem is more important than the method of solving it (Csikszentmihalyi, 2015).

To redefine startup problems, we analyzed discovery stage results through content analysis, focusing on the design ladder and key factors of design thinking (human, business, and technology). This helped answer RQ1 about why startups may offer products or services misaligned with actual market needs (Table 11).

Develop stage results

After redefining startup challenges from a design perspective and obtaining manager approval, we devised solutions for each startup's main problem. In the develop sessions, tailored consultations, utilizing discover and define stages results, and focused group meetings were conducted to reach the desired situation. Focused groups' data was analysed by content analysis method (Table 12); so data was coded to help us to deliver the solution.

Table 9 Degree of attention to business factors of business development and the level of design in startups

Startups	A	B	C	D	E
Existing design level (design ladder)	Lack of design	Style	Lack of design	Style	Lack of design
Paying attention to the factors of business development	Technology only (neglecting humans and business)	Business and technology (neglecting human)	Business and technology (neglecting human)	Human only (neglecting business and technology)	Business and technology (neglecting human)

Table 10 Codes extracted from discovery stage

No.	Codes	Description
1	Concern	The concern code represents the initial apprehensions of startups, which may not align with the actual problems of the market. Startups sometimes commence their operations based on personal preoccupations without sufficient research, leading to the development of solutions that do not resonate with the actual needs of the users
2	Problem	Startups mistake their concerns for the real issues without proper discovery, potentially leading to addressing the wrong challenges
3	Solution	Solution should be created after passing discover and define stage. Startups would deviate from normal path and start from solutions based on initial misunderstandings of problems, leading to potentially ineffective responses to the actual market needs
4	Mental backgrounds	The preconceived notions and assumptions that startup teams have influence their approach to problem discovery and definition, potentially leading them to address problems through a biased lens rather than objective analysis
5	Lack of user study	Startups tend to overlook the involvement of users in the initial problem-discovery stage. This leads to a failure in understanding user needs accurately, as startups often substitute their own assumptions for user perspectives
6	Starting from solution	Startups begin by proposing solutions before fully understanding the problem, potentially leading to a mismatch between their offerings and the actual needs of the market or users
7	Human value	Startups neglect to consider core human values and ethical considerations in their product design and business model, which could result in solutions that do not resonate with or even alienate their intended users
8	Technology	Startups often either focus heavily on the technological aspects of their products or services or neglect them, leading to a lack of viability or relevance to users' needs
9	Business model	Startups either have a one-dimensional approach to their business models, looking at them strictly from a financial perspective, or they altogether disregard the development of a robust business model. This lack of a balanced and comprehensive business strategy can lead to challenges in sustainability and scalability
10	Prejudice	Startups carry preconceived notions or biases that can influence their decision-making process, product development, and market strategies. These biases could potentially blind them to broader opportunities, and diverse customer needs, and can lead to under-representation or misalignment with the target audience they aim to serve
11	Efficiency and effectiveness	Efficiency and Effectiveness code implies that startups may struggle with optimizing their operations and processes, possibly due to limited resources, lack of expertise, or poor planning. This can result in suboptimal performance, where the company expends more effort, time, or resources than necessary to achieve its goals, leading to decreased productivity and potential difficulties in staying competitive in the market

Table 11 Define stage results

	Startup A	Startup B	Startup C	Startup D	Startup E
Redefining the problem (with design thinking method)	The founder's focus is on producing a solution to an undefined problem	Poor Product—Website Error (Graphics, Text)—High Cognitive Load (CLT)	Incomplete product	Sophisticated product design and packaging slow down the production process	High customer acquisition—Inability to retain a customer, inefficient internal manager
Desired design level (design ladder)	Strategy	Process	Style and strategy	Process	Process and strategy

Table 12 Codes extracted from develop stage

	Startup A	Startup B	Startup C	Startup D	Startup E
Solution space codes	1. Concerns need to be extracted from problem space	1. Concerns need to be extracted from problem	1. Concerns need to be extracted from problem	1. Concerns need to be extracted from problem stage	1. Concerns need to be extracted from problem stage
	2. Starting from problem space is needed	2. Improving technological aspect of product	2. Improving Human aspect of product	2. Improving technological aspect of product	2. Improving Human aspect
	3. Activates need to be aligned with the real problem	3. Improving Human aspect	3. Business model is needed	3. Improving Human aspect	3. Need to Judge real problem
	4. Need to Judge real problem			4. Business model is needed	4. Think out of box
	5. Think out of box				

Table 13 Develop stage results

	Solution creation (with design thinking method)
Startup A	Explain the founder about the principles of business design (participate in training courses, study specific resources appropriate to the current level of the founder, continue consulting sessions with the author), refer to the accelerator
Startup B	UX modification by Front-End specialist and content production specialist as project or hiring specialist
Startup C	Product completion (application) by the startup technical team or outsourcing the entire application design process
Startup D	Applying or training a designer with a stronger technical perspective, familiar with production methods, pivoting on the value proposition of the business model canvas, outsourcing design rather than outsourcing production
Startup E	Hiring a capable executive manager, hiring a financial manager, modifying the sales process, training organizational behavior to sales personnel, relocating the sales office, modifying the organization's processes (sales, warehousing, finance)

The solutions were presented based on the startup manager's priorities in terms of implementation capability, cost, and speed. These solutions are listed separately for each startup in Table 13.

Deliver stage results

At this stage, solutions for the real problems defined in previous stage were created and accepted by managers in terms of implementation capability. The purpose of this stage is to test and implement the solution and get feedback. The final solutions created in the development stage were delivered to managers for testing. The solutions were assessed and put into the test in accordance with the prioritizing set by the Analytical Hierarchy Process. Finally, after implementing the solutions, getting feedback, and modifying them, the final solution was selected (Table 14).

At deliver stage, the created solutions at stage 3 of DDF were tested and put into practice. After receiving feedback, the final solution was selected (Table 15).

Table 14 Deliver stage results

The final solution	
Startup A	Start designing a business model, getting advice from experienced mentors, making a presentation to an investor, attending startup events, attending an accelerator
Startup B	Modify the content of the website to reduce the cognitive load by the startup manager and with the weekly advice of the author
Startup C	The startup did not continue its cooperation due to internal problems. The consultation sessions ended
Startup D	Minimalist designs should be bought from outside designers
Startup E	A service designer should be hired and the sales process should be designed using service design tools

Table 15 Results of the implementation of the final solutions

The final solution	
Startup A	Completion of a logical business model, deployment in the business accelerator
Startup B	Growth of interaction rate by + 3/day after 1 month of implementation of the solution
Startup D	No measurable results have been reported so far
Startup E	Settlement of part of external debt, payment of part of arrears of personnel, reaching income of about $2000 monthly after 3 months of redesigning the sales process

Table 16 Results of implementing design thinking in startup

Startup A	Design thinking performance in problem space (stages 1 and 2)	Redefining the problem and achieving the right problem
	Design thinking performance in solution space (stages 3 and 4)	Creating a solution based on personal skills development
Startup B	Design thinking performance in problem space (stages 1 and 2)	Redefining the problem and get to the right problem
	Design thinking performance in solution space (stages 3 and 4)	Create a solution based on website UX modification
Startup C	Design thinking performance in problem space (stages 1 and 2)	Redefining the problem and getting to the right problem
	Design thinking performance in solution space (stages 3 and 4)	Creating a solution based on a user-centred design
Startup D	Design thinking performance in problem space (stages 1 and 2)	Redefining the problem and getting to the right problem
	Design thinking performance in solution space (stages 3 and 4)	Creating a solution based on the DFM approach
Startup E	Design thinking performance in problem space (stages 1 and 2)	Redefining the problem and getting to the right problem
	Design thinking performance in solution space (stages 3 and 4)	Designing the sales process using service design tools

The results of implementing the design thinking approach to solve startup problems, listed in Table 12, show that the main problem of the startups analysed in this research is "mis-defining the startup problem (Table 16)".

Our results, measured by the RG over 3 months, indicate significant improvements in the financial performance of the startups post-implementation of design thinking. Each startup showed a marked increase in revenue, as illustrated in Table 17,

demonstrating the efficacy of design thinking in enhancing business outcomes within a short timeframe.

Discussion

Perceived vs. actual startup challenges

Our analysis has unearthed a profound disparity between the perceived problems voiced by startup founders and those genuinely afflicting their ventures. For example, startup A, within the hairstyling domain, put forth a discount card program aiming to bolster annual profits. Startup B, operating as a virtual social forum, cited new member shortages as its impediment. Startup C, in mobile car repair, ascribed its difficulties to weak brand visibility. Startup D, a promotional gifts design and consultancy firm, identified production delays as its central obstacle. Meanwhile, startup E associated its stunted growth with sluggish sales.

Nonetheless, a meticulous Discover stage, comprising interviews and observations, disclosed misalignments between these professed concerns and the foundational issues leads present products or services that do not match the market needs that has answered to RQ1. For startup A, the genuine quandary lay not in the absence of discounts but in the founder's haste towards solution generation without apt problem definition. Challenges for startup B transcended mere numerical user growth, rooted instead in a deficient product plagued by website flaws, substandard graphics, and an overwhelming user experience. Startup C grappled not just with obscurity but with an inherently flawed product. Similarly, startup D faced not mere production lags but a complex design and packaging process that was inherently time-intensive. Startup E's challenge was broader than lackluster sales; it was facing high customer turnover coupled with inefficient internal management.

The core findings of this study underscore the paramount importance of accurately defining problems within startups through design thinking covering RQ2 and RQ3. Misalignments between startups' offerings and market demands were predicted to be prevalent factors for failure, in line with Eisenmann (2021) and Sanasi et al. (2023). Yet, our explorations unearthed a deeper genesis of failure: erroneous problem diagnosis. This notion adds a fresh perspective to the established understanding of startup complexities by Crnogaj and Rus (2023), who concentrated on external market forces, by revealing the substantial impact of internal processes and entrepreneurial mindsets on startup success. Particularly unexpected was the emphasis on technological and business elements to the neglect of the human factor, underscoring the intricate

Table 17 Revenue growth over a 3-month period before and after design thinking implementation

	Pre-design thinking 3-month revenue	Post-design thinking 3-month revenue
Startup A	$5230	$15,230
Startup B	$3120	$12,650
Startup C	$4310	$6520
Startup D	$2180	$5430
Startup E	$1570	$20,560

nature of design thinking, embracing the triad, and echoing Arnold's (1959) business development model that integrates human values with technology and business acumen.

Our findings show that design thinking improves alignment between startup offerings and market needs, supporting Open Innovation 2.0 principles (Curley & Salmelin, 2017). This helps startups meet demands by iterating based on user feedback. The study also highlights the complexity of misaligned perceived and actual problems, aligning with Complex Systems and Innovation Theory (Uhl-Bien & Arena, 2017). In addition, Change Management in Innovative Organizations supports using design thinking to drive significant changes in processes and culture, demonstrating its effectiveness for long-term success (Anderson & Anderson, 2010).

Our outcomes resonate with Rittel and Webber's (1973) principles on wicked problems and mirror Camillus's (2008) insights on business convolutions, illustrating startups' tendencies to mislabel wicked issues as tame, inadvertently steering efforts astray. Absent engagement in the problem space, startups revert to a solution-centric modality, which falters against wicked problems, epitomized by the elevated failure rates detailed by Griva et al. (2023).

The results related to our hypotheses suggest that design thinking plays a crucial role in enhancing the problem-solving capabilities of startups. Hypothesis 1 was strongly supported, indicating that design thinking aids in the accurate identification of wicked problems. However, Hypothesis 2 and Hypothesis 3, while supported, suggest that the extent of impact can vary based on other factors, such as the startup's stage and industry. These findings contribute to the broader discourse on the effectiveness of design thinking in complex environments.

The efficacy of design thinking in cultivating a problem space exploration ethos is conspicuous in our findings, where startups immersed in a reconceptualized process demonstrated enhanced market alignment, a testament to the explorative essence of DDF by separating problem and solution space; and that is how RtD can aid in identifying and articulating these misalignments (RQ2). The misdefinition of problems by startups stems from key factors:

- Commencing the problem-solving venture by homing in on solutions while overlooking or misunderstanding the problem space.
- Neglect of thorough problem definition, overlooking stages 1 and 2 of the design thinking process.
- Inattention to at least one of the three factors of business development: human, business, or technology.
- Conflating concerns (Double Diamond's first stage) with actual problems (second stage).
- Fervor in generating correct solutions for misjudged problems.
- Oversimplifying wicked problems and approaching them as if they were tame.

Essentially, the crux of the examined startups' issues lies in the misinterpretation of what constitutes the problem. Factors contributing to such misdefinitions can be attributed to:

- An oversight of the three factors of business development.
- Drawing parallels between wicked and tame issues.
- Disregard for the problem space.

Design thinking, as explored in RQ3, reveals the nuanced and complex nature of wicked problems startups face. Startups with teams rich in technical expertise, possessing analytical thought rooted in logic, often find their focus on elaborating and developing detailed solutions, yet may minimize the importance of the accuracy of problem space identification. Such teams tend to fixate on the 'technological' and 'business' factors of the innovation triad, occasionally sidelining the 'human' component.

Implications for policy and practice in unfamiliar ecosystems

The results underscore the necessity for startups to engage deeply with the problem space before jumping to solutions. This engagement requires a comprehensive understanding and application of design thinking principles, emphasizing empathy, iterative learning, and a balance among the three factors of business development. Such an approach can help startups navigate the complex, wicked problems they face by ensuring their solutions are more aligned with market needs and human values. For policy, there is a clear indication that supporting structures, such as educational programs, incubators, and accelerators, should focus not only on the technical aspects of startup development but also on fostering a design thinking mindset. The integration of design thinking into entrepreneurial ecosystems plays a crucial role in fostering sustainable business innovation, particularly in unfamiliar and complex environments. This methodology, which emphasizes empathy, ideation, prototyping, and testing (Sadiku et al., 2019), is not only vital for startup success but also serves as a strategic tool for policymakers aiming to create more resilient ecosystems. However, despite its recognized value, design thinking remains ambiguously defined in many contexts, highlighting the need for clearer policy frameworks and structured educational programs that enhance its applicability (Kwon et al., 2024). The ability to navigate uncertainty is particularly critical in unfamiliar ecosystems, where conventional business strategies often fall short. As Vnukova et al. (2022) argue, design thinking enhances startup success by ensuring a more human-centered approach, which is especially beneficial in dynamic, high-risk environments.

Incubators and accelerators, as key enablers of entrepreneurial growth, should incorporate design thinking methodologies into their curricula to improve problem-solving capabilities, increase market alignment, and enhance long-term business viability (Ayatse et al., 2017; Lange & Johnston, 2020). By providing a structured approach to innovation, design thinking enables startups to iteratively refine their products and services based on real user needs, thereby increasing their predictability of success (Dell'Era et al., 2020). This is particularly relevant in emerging markets, where rapid technological changes and socio-economic factors contribute to heightened business complexity. Furthermore, the combination of design thinking with lean startup methodologies can significantly improve the efficiency of product development and market readiness (Dobrigkeit et al., 2020), allowing startups to better adapt to unfamiliar environments. From a policy perspective, fostering a design-thinking mindset within entrepreneurial ecosystems can enhance problem articulation and decision-making at both micro and

macro levels. The gap between education and workforce demands, as highlighted by Tickoo and Grammer (2022), necessitates the integration of design thinking principles into academic and professional training programs to equip entrepreneurs with the necessary skills to tackle wicked problems. In addition, Carlgren et al., 2016a, 2016b, 2016c) reinforce the role of design thinking in fostering innovation across various sectors, including policy-making, by promoting structured yet flexible approaches to complex challenges. By embedding design thinking into both startup support structures and policy initiatives, ecosystems can be better equipped to handle the uncertainties and ambiguities inherent in unfamiliar business landscapes. These findings underscore the importance of shifting policy priorities toward design-driven innovation ecosystems, where adaptive problem-solving frameworks are embedded into regulatory and support structures. Governments and economic development agencies should incentivize the adoption of design thinking within startup incubators, offering grants or tax benefits for programs that integrate iterative and human-centered design methodologies. In addition, cross-sector collaboration between academia, industry, and policymakers can further enhance the diffusion of design thinking as a scalable approach to navigating complexity in uncertain business environments. These policy interventions should be complemented by cultural and institutional shifts that emphasize problem definition as a foundational step in business strategy. Policies could encourage the integration of design thinking courses within entrepreneurial education, provide funding for design thinking workshops, and incentivize mentorship programs that connect startups with design thinking practitioners.

In practice, startups should be encouraged to adopt a more user-centric approach from the outset. This involves investing time and resources in understanding the user's needs, desires, and contexts through empathetic engagement and research. Furthermore, startups should be guided to embrace iterative development processes, where feedback loops with real users are integral to refining and validating their solutions. The implications extend to the broader startup ecosystem, suggesting a need for a cultural shift towards valuing deep problem understanding as much as innovative solution development. This shift can be facilitated by creating more opportunities for cross-disciplinary collaboration, where designers, technologists, and business expert's work together to uncover and address the underlying complexities of wicked problems. The findings of this study underscore the potential for policy-driven support to significantly enhance the success rates of startups in regions similar to ours. By aligning startup needs with specific policy measures, such as financial incentives and regulatory support, policymakers can directly influence the entrepreneurial landscape. This research suggests that a strategic focus on supporting innovation through comprehensive policy frameworks can lead to sustainable growth in the startup ecosystem.

Conclusion

This research not only highlights the challenges faced by startups in Tehran but also provides actionable insights for policymakers on how to effectively support these entities. The practical application of these insights can lead to a more vibrant and resilient startup ecosystem, which is crucial for economic growth and innovation in emerging markets.

Our research culminates in bridging a notable schism within prior studies, pivoting the lens towards the intricate machinery operating within startups, specifically, the pivotal processes of problem articulation and examination. It casts the spotlight on the theoretical significance of distinguishing wicked from tame problems, as defined by Rittel and Weber (1973), advocating for a pivot to a truly human-centric modus operandi, as accentuated by Arnold (1959).

Our research confirms the beneficial impact of design thinking on managing wicked problems in startups. By testing our hypotheses, we have added empirical support to the assertion that design thinking not only aids in the accurate identification of problems but also enhances the overall success and market alignment of startups. These insights underscore the value of incorporating design thinking into the strategic frameworks of startups navigating complex challenges.

This study further provides theoretical credence to the DDF, proposed by Gustafsson (2019), validating its application as a navigational tool through the turbulent seas of startup complexities and emphasizing the integration of design thinking tenets into the sphere of entrepreneurial pedagogy whether for enterprises or startups which is being studied recently in different researches, such as Garrel and Jahn (2022).

The comparative analysis of RG also clearly demonstrates that the application of design thinking has had a significant positive impact on the performance of all five startups. Therefore, this approach can be introduced as a powerful tool for enhancing company performance.

For entrepreneurs and startup founders, the narrative of this study underscores the strategic imperative to allocate time and resources judiciously for the meticulous definition of challenges before the inception of solution ideation. We endorse design thinking beyond its conventional guise of a creative avenue, framing it as an orderly paradigm for problem-solving that harmoniously weaves empathy with technological savvy and commercial insight. This narrative also extends an invitation to policymakers to cultivate entrepreneurial ecosystems, where design thinking flourishes, leaning perhaps on the instrument of subsidized educational initiatives that promulgate these values. Our inquiry highlights three distinct business evolution factors that are pivotal in accurately delineating problems:

1. Insufficient focus on factors of business development

   The explicit recognition of pivotal factors within business development, namely human, technological, and commercial, illuminates the problem scope for startups, enhancing its resolution and precision. The human element, though central in problem conceptualization, finds its complementary forces in technology and business within the realm of problem solutions. Empathy emerges as a non-negotiable in crafting superior solutions, given the innately human-centric ethos of design endeavours. A deficiency in empathy could result in a myopic understanding of user demands, thereby weakening the human aspect of the investigative and creative process.
2. Misconstruing wicked problems as tame

Star-ups, often unversed in the nuances of wicked problems, tend to miscast all challenges as tame, overlooking the need for extensive exploration within the problem domain. Wicked problems, shrouded in solution and definition ambiguity, necessitate a thorough analysis and the establishment of an 'exploration space', a concept well-articulated in design thinking frameworks but frequently overlooked by star-ups.

3. Neglecting the problem space
   The singular, most significant predictor of startup failure could well be the oversight of the 'problem space.' Star-ups evidenced neither a historical engagement with problem space exploration, pivotal in the initial stages of the DDF or alternative design thinking models nor demonstrated any inclination to do so, often dismissing it as futile. Only after targeted training and advisory sessions did founders recognize the value of scrutinizing the problem space. Absent baseline training on wicked problem methodologies and devoid of team members with such expertise, startups venture into a domain rife with uncertainty and convoluted systems. The obscurity in both problem (what) and solution (how) diagnostics underscores the potency of focusing on the problem space over the solution space for startup teams. Startups might exhibit competence in devising plausible solutions, occasionally even excellent solutions to the incorrect problem. The three pillars of business development hold substantial weight in entrepreneurial success, yet startups engage with these pillars only superficially, often to their detriment and potential downfall.

In navigating the intricate terrain of unfamiliar ecosystems, our exploration underscores the pivotal role of design thinking in steering startups towards addressing wicked problems effectively. The implications for policy and practice, as elucidated in our discussion and literature review, highlight the necessity for a paradigm shift towards an integrated, human-centric approach. For policymakers, fostering an environment that champions design thinking methodologies could significantly enhance startup resilience and adaptability. This entails investing in educational programs that prioritize design thinking skills, encouraging partnerships between startups and design thinking experts, and providing platforms for knowledge exchange on best practices. Practically, startups must imbibe a culture of empathy, iterative learning, and multidisciplinary collaboration to unveil and tackle the underlying complexities of wicked problems. Embracing these strategies not only aids in aligning startups' offerings with market needs but also in cultivating innovative solutions that are sustainable and impactful. Thus, our findings advocate for a concerted effort among stakeholders to embed design thinking at the core of policy formulation and entrepreneurial practice, ensuring startups are well-equipped to thrive in the face of uncertainties inherent in unfamiliar ecosystems.

Limitations and future research

Future research avenues present the opportunity to unravel the influence of design thinking education on mitigating startup failure rates. Deliberate comparative studies assessing the outcomes of startups thoroughly immersed in design thinking against those with minimal integration could provide concrete evidence bolstering the theoretical propositions of this study. In addition, scrutiny into the efficiency of distinct

design thinking frameworks within varied cultural and economic landscapes could yield insights into their universal applicability and relevance. The exploration of digital transformation's role in augmenting startups' design thinking proficiencies offers a rich domain for inquiry, particularly pertinent in the context of our increasingly digital global economy.

While design thinking stands as a potent and insightful approach for problem comprehension and innovative solution formulation, it merits noting that this paradigm is in an evolutionary phase due to its relatively recent inception and has yet to reach full maturity. Hence, precision and contemporaneity in the deployment of this methodology are critical to ensure that the broader tenets of design are aptly tailored to the exigencies of the business milieu.

Abbreviations

RtD	Research through design
DDF	Double diamond framework
SWOT	Strengths, weaknesses, opportunities, and threats
RG	Revenue growth
CLT	Cognitive load theory
OOH	Out-of-home advertising
DFM	Design for manufacturing

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